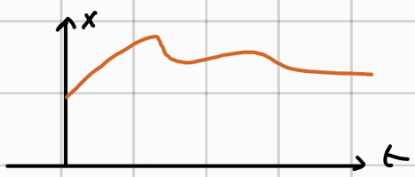
State Space

2nd order D.E. $\xrightarrow{S.S.}$ 2 first order D.E

System of $\left\{\begin{array}{l} 2^{nd} \text{ order} \\ \text{equations} \\ (n) \end{array}\right.$ $\xrightarrow{S.S.}$ 2n first order D.E.

Example: $\ddot{x} + \alpha_1 \dot{x} + \alpha_2 x = u$



x : output 2nd order D.E.

u : input

$x = y_1$ $\quad \dot{y}_1 = \dot{x} = y_2$

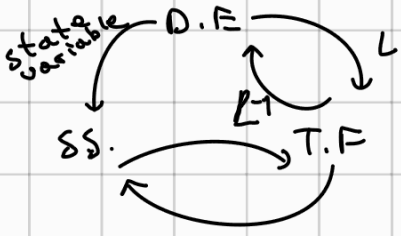
$\dot{x} = y_2$ $\quad \dot{y}_2 = \ddot{x} = u - \alpha_1 \dot{x} - \alpha_2 x$

$= u - \alpha_1 y_2 - \alpha_2 y_1$

state variables

$$\frac{d}{dt}\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\alpha_2 & -\alpha_1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

state variable vector (x) — A — X — input vector (B) — u output



$$\dot{\vec{X}} = A\vec{X} + \vec{B}u$$

State space representation

↳ Fundamental operation (Mathematically) is two lower the order of D.E to half. If you are having 2 n equation of 2nd order this method produces 4n D.E. of 1st ord.

* States of a system are the min number of variable to be known at t=to in order to describe the dynamics (response) at any later time.

Ex $\dddot{y} + \alpha_1 \ddot{y} + \alpha_2 \dot{y} = \ddot{u} + \alpha_3 u$, y: output, u: input

States can be $y(0), \dot{y}(0) \rightarrow y(t) = \checkmark$ //

variables → output
→ input
→ state

State space representations

- physical variable ✓
- phase variable ✓
- canonical variable
  - observer ✗
  - controller ✗

In physical variable form we just look at the energy storing elements

Capacitor: $\frac{1}{2}CV^2$ — Voltage: State variable

Inductor: $\frac{1}{2}Li^2$ — Current //

mass: $\frac{1}{2}I\omega^2 / \frac{1}{2}mv^2$ — ω / v : state variable

spring: $\frac{1}{2}kx^2$ //

Phase variable form

Case 1: No derivative of the input

$\dot{u} = \frac{du}{dt}$

- forward difference $\lim_{\Delta t \to 0} \frac{u(t+\Delta t) - u(t)}{\Delta t}\Big|_{t=0}$
- backward difference $\lim_{\Delta t \to 0} \frac{u(t) - u(t-\Delta t)}{\Delta t}$ $t = -\Delta t$
- central difference $\lim_{\Delta t \to 0} \frac{u(t+\frac{\Delta t}{2}) - u(t-\frac{\Delta t}{2})}{\Delta t}$

$\dot{y} = u$ ⇓ y: output, u: input

$\ddot{y} = \ddot{u}$ y: output, u: input

for $\ddot{y} = \dot{u} - \dot{y}$ you have to know $y(0), \dot{y}(0)$ &. and $\dot{u}$ is given

---

$$y^{(n)} + a_1 y^{(n-1)} + a_2 y^{(n-2)} + \cdots + a_n y = u$$

(time over)

state variables:

$x_1 = y$

$x_2 = \dot{y}$

$x_3 = \ddot{y}$

$\vdots$

$x_n = y^{(n-1)}$

$\xrightarrow{\frac{d}{dt}}$

$\dot{x}_1 = \dot{y} = x_2$

$\dot{x}_2 = \ddot{y} = x_3$

$\vdots$

$\dot{x}_n = y^{(n)} = u - a_1 y^{(n-1)} - a_2 y^{(n-2)} - \cdots - a_n y$ (x_n, x_{n-1}, x_1)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ 0 & 0 & 0 & 1 & \cdots \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} u$$

$$\dot{\vec{X}} = \underset{n\times n}{A}\ \underset{n\times 1}{\vec{X}} + \underset{n\times 1}{B}u$$

$\dot{\vec{X}} = A\vec{X} + Bu \rightarrow$ state space representation

A: state matrix

B: input vector

C: output vector

D: Direct-transport matrix/vector

$$y_{1\times 1} = \underbrace{[1\ 0\ 0\ \cdots\ 0]}_{C} \underbrace{\begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{Bmatrix}}_{X} = C\vec{x} + \cancel{Du}^{\,0}$$

Case 2: Derivative of input is present

$$\underbrace{\dddot{y} + a_1\ddot{y} + a_2\dot{y} + a_3 y}_{\text{LHS}} = \underbrace{b_0\ddot{u} + b_1\dot{u} + b_2 u + b_3 u}_{\text{RHS}}$$

→ statically related

(can be chosen)

u: input

y: output

$$\begin{cases} x_1 = y \\ x_2 = \dot{y} \\ x_3 = \ddot{y} \end{cases} \longrightarrow \begin{cases} \dot{x}_1 = \dot{y} = x_2 \\ \dot{x}_2 = \ddot{y} = x_3 \\ \dot{x}_3 = \dddot{y} = \text{RHS} - \text{LHS} \end{cases}$$

LHS: left hand side

RHS: Right hand side

Burdan SS. formatına sokamayız

O güzden bunu yapalım

$$y^{(n)} + a_1 y^{(n-1)} + a_2 y^{(n-2)} + \cdots + a_n y = b_0 u^{(n)} + b_1 u^{n-1} + \cdots + b_n u$$

u: input

y: output

number of state variables n

→ ezberlemeye gerek yok gerekirse sınavda verilir

$$\begin{cases} x_1 = y - \beta_0 u \\ x_2 = \dot{y} - \beta_0 \dot{u} - \beta_1 u = \dot{x}_1 - \beta_1 u \\ x_3 = \ddot{y} - \beta_0 \ddot{u} - \beta_1 \dot{u} - \beta_2 u = \dot{x}_2 - \beta_2 u \\ \vdots \\ \dot{x}_n = \dot{x}_{n-1} - \beta_{n-1} u \end{cases}$$

$$\beta_0 = b_0$$

$$\beta_1 = b_1 - a_1\beta_0$$

$$\beta_2 = b_2 - a_1\beta_1 - a_2\beta_0$$

$$\vdots$$

$$\beta_{n-1} = b_{n-1} - a_1\beta_{n-2} - \cdots - a_{n-2}\beta_1 - a_{n-1}\beta_0$$

Phase variable form

$$\begin{cases} \dot{x}_1 = \dot{y} - \beta_0 \dot{u} = x_2 + \beta_1 u \\ \dot{x}_2 = \ddot{y} - \beta_0 \ddot{u} - \beta_1 \dot{u} = x_3 + \beta_2 u \\ \vdots \\ \dot{x}_n = -a_n x_1 - a_{n-1} x_2 - \cdots - a_1 x_n + \beta_n u \end{cases}$$

$$\begin{Bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{Bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 0 & \cdots & 0 \\ \vdots & & & & & \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix} u$$

$$\dot{\vec{x}} = A\vec{x} + Bu$$

$$y = x_1 + \beta_0 u = \underbrace{[1\ 0\ 0 \cdots 0]}_{C} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + (\beta_0)\, u$$

$$y = C\vec{x} + Du$$

$$T.F = \frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n}$$

Örnek

$$\underbrace{\dddot{y} + a_1 \ddot{y} + a_2 \dot{y} + a_3 y}_{LHS} = \underbrace{b_0 \ddot{u} + b_1 \dot{u} + b_2 \dot{u} + b_3 u}_{RHS}$$

$\beta_0 = b_0$

$\beta_1 = b_1 - a_1 \beta_0$

$\beta_2 = b_2 - a_1 \beta_1 - a_2 \beta_0$

$\beta_3 = b_3 - a_1 \beta_2 - a_2 \beta_1 - a_3 \beta_0$

$x_1 = y - b_0 u$

$x_2 = \dot{y} - b_0 \dot{u} - (b_1 - a_1 \beta_0) u$

$x_3 = \ddot{y} - b_0 \ddot{u} - (b_1 - a_1 \beta_0) \dot{u} - (b_2 - a_1 \beta_1 - a_2 \beta_0) u$

$$\begin{cases} x_1 = y - \beta_0 u \\ x_2 = \dot{y} - \beta_0 \dot{u} - \beta_1 u \\ x_3 = \ddot{y} - \beta_0 \ddot{u} - \beta_1 \dot{u} - \beta_2 u \end{cases} \longrightarrow \begin{cases} \dot{x}_1 = \dot{y} - \beta_0 \dot{u} = x_2 + \beta_1 u \\ \dot{x}_2 = \ddot{y} - \beta_0 \ddot{u} - \beta_1 \dot{u} = x_3 + \beta_2 u \\ \dot{x}_3 = \dddot{y} - \beta_0 \dddot{u} - \beta_1 \ddot{u} - \beta_2 \dot{u} \end{cases}$$

$$\beta_0 = b_0$$

$$\beta_1 = b_1 - a_1\beta_0$$

$$\beta_2 = b_2 - a_1\beta_1 - a_2\beta_0$$

$$\dddot{y} = -a_1\ddot{y} - a_2\dot{y} - a_3 y + b_0\dddot{u} + b_1\ddot{u} + b_2\dot{u} + b_3 u$$

$$= -a_1(\ddot{y} - \beta_0\ddot{u} - \beta_1\dot{u} - \beta_2 u) - a_1\beta_0\ddot{u} - a_1\beta_1\dot{u} - a_1\beta_2 u$$

$$- a_2(\dot{y} - \beta_0\dot{u} - \beta_1 u) - a_2\beta_0\dot{u} - a_2\beta_1 u$$

$$- a_3(y - \beta_0 u) - a_3\beta_0 u + b_0\dddot{u} + b_1\ddot{u} + b_2\dot{u} + b_3 u - \beta_0\dddot{u} - \beta_1\ddot{u} - \beta_2\dot{u}$$

$$= -a_1x_3 - a_2x_2 - a_3x_1 + \ddot{u}(b_0 - \beta_0) + \dot{u}(b_1 - \beta_1 - a_1\beta_0)$$

$$+ \dot{u}(-a_1\beta_1 - a_2\beta_0 + b_2 - \beta_2) + u(-a_1\beta_2 - a_2\beta_1 - a_3\beta_0 + b_3$$

$(b_0 - \beta_0) = 0$, $(b_1 - \beta_1 - a_1\beta_0) = 0$, $(-a_1\beta_1 - a_2\beta_0 + b_2 - \beta_2) = 0$, $(-a_1\beta_2 - a_2\beta_1 - a_3\beta_0 + b_3) = \beta_3$

$$\begin{cases} \dot{x}_1 = x_2 + \beta_1 u \\ \dot{x}_2 = x_3 + \beta_2 u \\ \dot{x}_3 = -a_1x_3 - a_2x_2 - a_3x_1 + \beta_3 u \end{cases} \qquad y = x_1 + \beta_0 u$$

$$\dot{\vec{X}} = A\vec{x} + Bu \quad \rightarrow \text{time domain}$$

$$\dot{y} = C\vec{X} + Du$$

$\mathcal{L}$ (S.S $\rightarrow$ TF)

$$sX(s) - X(0) = AX(s) + BU(s) \rightarrow (sI - A)\vec{X}(s) = BU(s)$$

$$\downarrow$$

$$X(s) = (sI - A)^{-1}BU(s)$$

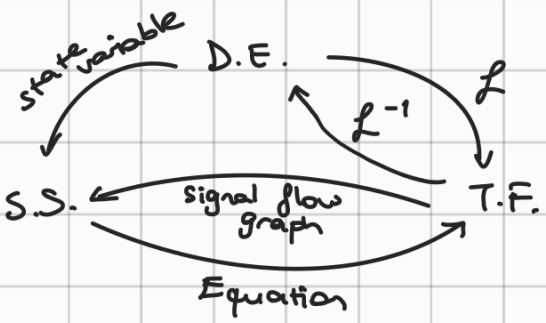
$$Y(s) = CX(s) + PU(s)$$

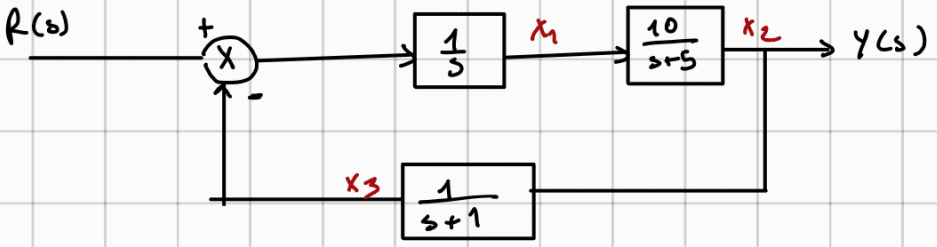
$$Y(s) = C(sI - A)^{-1}BU(s) + DU(s)$$

$$G(s) = \frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D$$

↳ a vector of TF

$$G(s) = \frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D$$





$\frac{Y(s)}{X(s)} = \frac{1}{s}$ → integration

$\frac{Y(s)}{X(s)} = s$ → derivative

$\frac{x_3}{x_2} = \frac{1}{s+1}$, $\frac{x_2}{x_1} = \frac{10}{s+5}$, $\frac{x_1}{R-x_3} = \frac{1}{s}$

$x_2 = s x_3 + x_3$

$\Rightarrow s x_3 = x_2 - x_3$

$\Rightarrow \dot{x}_3 = x_2 - x_3$

$10 x_1 = s x_2 + 5 x_2$

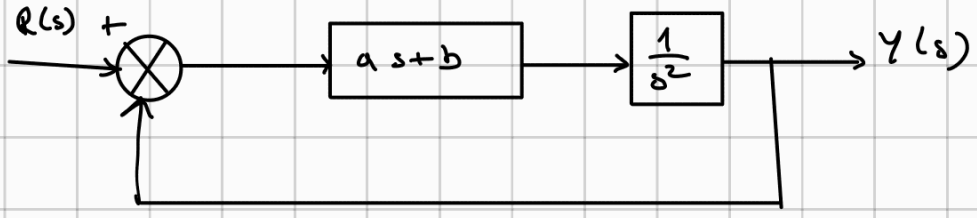
$\Rightarrow s x_2 = 10 x_1 - 5 x_2$

$\Rightarrow \dot{x}_2 = 10 x_2 - 5 x_2$

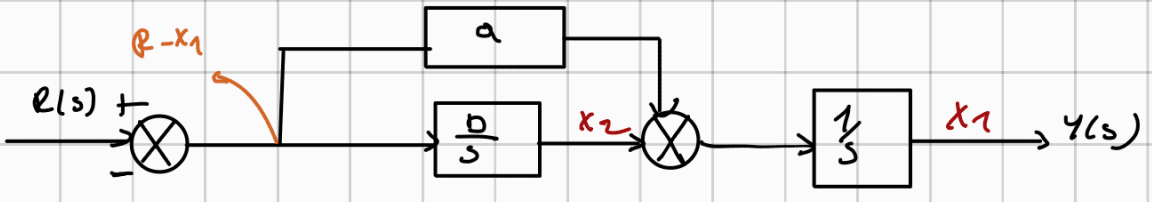
$s x_1 = R - x_3$

$\Rightarrow \dot{x}_1 = r - x_3$

$$\left.\begin{aligned} \dot{x}_1 &= -x_3 + r \\ \dot{x}_2 &= 10 x_1 - 5 x_2 \\ \dot{x}_3 &= x_2 - x_3 \end{aligned}\right\} \rightarrow \begin{Bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{Bmatrix} = \begin{bmatrix} 0 & 0 & -1 \\ 10 & -5 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} + \begin{Bmatrix} 1 \\ 0 \\ 0 \end{Bmatrix} r$$



$$\frac{as+b}{s^2} = \frac{a}{s} + \frac{b}{s^2} = \frac{1}{s}\left(a + \frac{b}{s}\right)$$



$\frac{X_2}{R-x_1} = \frac{b}{s}$ (1) $\qquad s x_2 = b(R - x_1)$

$\frac{X_1}{X_2 + a(R-x_1)} = \frac{1}{s}$ (2) $\qquad s x_1 = x_2 + a(R - x_1)$

Ex: $m_1\ddot{y}_1 = b_1(\dot{y}_2 - \dot{y}_1) - k_1 y_1 + u_1$

$m_2\ddot{y}_2 = b_2(\dot{y}_2 - \dot{y}_1) - k_2 y_2 + u_2$

m_1, m_2: mass values

b_1: damping value

k_1, k_2: stiffness values

} these are all number

u_1, u_2: inputs, y_1, y_2: outputs

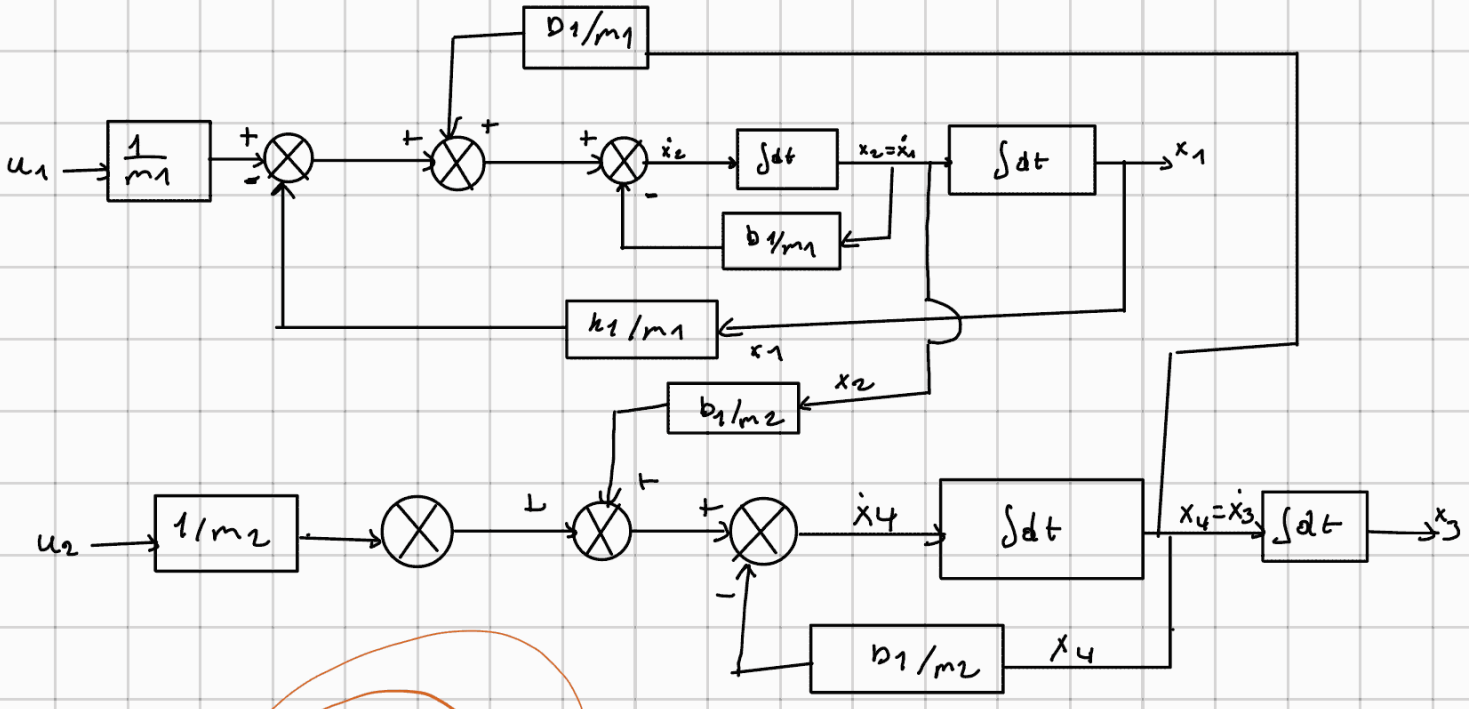
S.S.

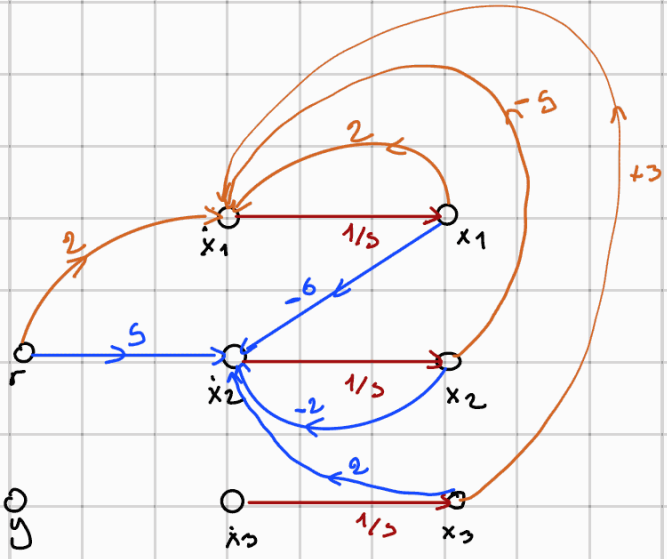
$x_1 = y_1$ $\quad \dot{x}_1 = \dot{y}_1 = x_2$

$x_2 = \dot{y}_1$ $\Rightarrow$ $\dot{x}_2 = \ddot{y}_1 = \frac{1}{m_1}\left[b_1(\dot{y}_2 - \dot{y}_1) - k_1 y_1 + u_1\right] = \frac{1}{m_1}\left[b_1(x_4 - x_2) - k_1 x_1 + u_1\right]$

$x_3 = y_2$ $\Rightarrow$ $\dot{x}_3 = \dot{y}_2 = x_4$

$x_4 = \dot{y}_2$ $\quad \dot{x}_4 = \ddot{y}_2 = \frac{1}{m_2}\left[-b_1(\dot{y}_2 - \dot{y}_1) - k_2 y_2 + u_2\right] = \frac{1}{m_2}\left[-b_1(x_4 - x_2) - k_2 x_3 + u_2\right]$





$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{-k_1}{m_1} & \frac{-b_1}{m_1} & 0 & \frac{b_1}{m_1} \\ 0 & 0 & 0 & 1 \\ 0 & \frac{-b_2}{m_2} & \frac{-k_2}{m_2} & \frac{-b_1}{m_2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 0 \\ \frac{1}{m_1} & 0 \\ 0 & 0 \\ 0 & \frac{1}{m_2} \end{bmatrix}}_{2\times 4} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$\dot{X}$ = A X + B $\vec{u}$

$\dot{x}_2 = b r - x_1 b$

$\dot{x}_1 = x_2 + a(r - x_1)$

$\dot{x}_2 = b R - x_1 b$

$\dot{x}_1 = x_2 + a(R - x_1)$

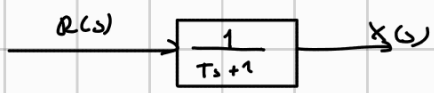
$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \underbrace{\begin{bmatrix} -a & 1 \\ -b & 0 \end{bmatrix}}_{2\times 2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} b \\ a \end{bmatrix} [r]$$

$$\dot{\vec{X}} = A\vec{X} + Bu$$

$$Y = Cx + Du$$

1st order system

III) Ramp Response



$r(t) = t$

$R(s) = \frac{1}{s^2}$

$X(s) = R(s)\frac{1}{Ts+1} = \frac{1}{s^2(Ts+1)}$

$s_{1,2} = 0$

$s_3 = -\frac{1}{T}$

$= \frac{A}{s^2} + \frac{B}{s} + \frac{C}{Ts+1}$

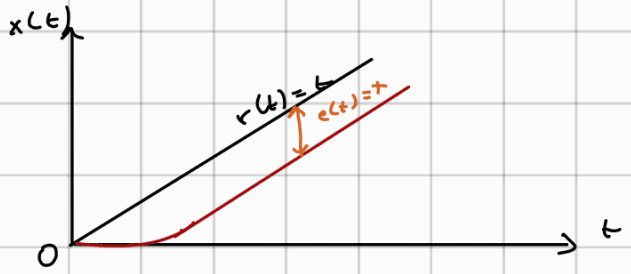
$A = \cancel{s^2}\left(\frac{1}{\cancel{s^2}(Ts+1)}\right)\Big|_{s=0} = 1$

$B = \frac{d}{dt}\left(s\frac{1}{s^2(Ts+1)}\right)\Big|_{s=0} = -T$

$C = (Ts+1)\frac{1}{s^2(Ts+1)}\Big|_{s=-\frac{1}{T}} = T^2$

$$X(s) = \frac{1}{s^2} - \frac{T}{s} + \frac{T^2}{Ts+1} = \frac{T}{s+\frac{1}{T}}$$

$$\mathcal{L}^{-1}: \quad x(t) = t - T + Te^{-t/T}$$



$$\hat{x} = 1 - e^{-t/T}$$

$$e(t) = r(t) - x(t)$$

$$= t - (t - T + Te^{-t/T})$$

$$= T - Te^{-t/T}$$

⊜ $t = 0 \longrightarrow e(0) = 0$

$t \to \infty \Rightarrow e(\infty) = e_{ss} = T$

Accuracy is decided by 'T'.

Ramp response = $\int$ unit response = $\iint$ impulse respone

$$\frac{d}{dt}(\text{ramp resp}) = \text{step resp}$$

$$\frac{d^2}{dt^2}(\text{ramp resp}) = \text{impulse resp}$$

2nd order response

RLC in series

$$\frac{V_c(s)}{V(s)} = \frac{1}{LCs^2 + RCs + 1} = \frac{1/LC}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

RLC in parallel

$$\frac{\Phi(s)}{I(s)} = \frac{1}{Cs^2 + \frac{s}{R} + \frac{1}{L}} = \frac{1/C}{s^2 + \frac{s}{RC} + \frac{1}{LC}}$$

Spring mass Damper System

$$\frac{X(s)}{F(s)} = \frac{1}{ms^2 + cs + k} = \frac{1/m}{s^2 + \frac{c}{m}s + \frac{k}{m}}$$

$$\Rightarrow \frac{\text{output}}{\text{input}} = \frac{X(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

ω_n = naturally freqency

ζ = zeta = damping ratio

no zeros but 2 poles are present

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

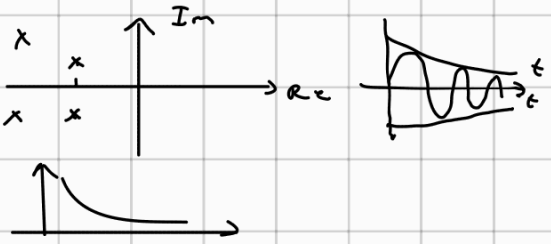
$$s_{1,2} = \frac{-2\zeta\omega_n \mp \sqrt{(2\zeta\omega_n)^2 - 4\omega_n^2}}{2}$$

$$= -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

$$= \omega_n\left(-\zeta \pm \sqrt{\zeta^2 - 1}\right)$$

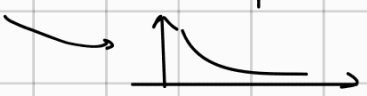
Underdamped Systems

if $\zeta < 1$ $\Rightarrow$ complex conjugate poles (always oscilatory)



critically damped systems

(zoning) if $\zeta = 1$ $\Rightarrow$ repeated reel valued poles



if $\zeta > 1$ $\Rightarrow$ seperate and real valued poles overdamped system

Homework (homogen solution)

Response due to external excitation (particular solution)

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Specify the input

I) Impulse excitation $R(s) = 1$

case I) $\zeta < 1$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{(s + \zeta\omega_n - i\omega_d)(s + \zeta\omega_n + i\omega_d)}$$

$$= \frac{A}{s + \zeta\omega_n - i\omega_d} + \frac{B}{s + \zeta\omega_n + i\omega_d}$$

$$s_{1,2} = +\omega_n\left(-\zeta \pm \sqrt{\zeta^2 - 1}\right)$$

$$= \omega_n\left(-\zeta \pm i\sqrt{1 - \zeta^2}\right)$$

$$= -\zeta\omega_n \pm i\,\omega_n\sqrt{1 - \zeta^2}$$

damped natural freq = ω_d

The response frequency for damped systems is ω_d

$$\frac{C(s)}{R(s)} = \frac{A}{s + \zeta\omega_n - i\omega_d} + \frac{B}{s + \zeta\omega_n + i\omega_d} \Rightarrow c(t)$$

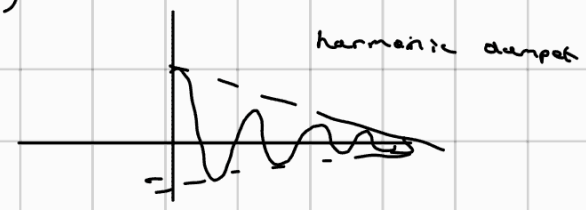
$$A = \frac{\omega_n^2}{s + \zeta\omega_n + i\omega_d}\Bigg|_{s = -\zeta\omega_n + i\omega_d}$$

$$B = \left(\frac{\omega_n^2}{s+\zeta\omega_n+i\omega_d} \right)_{s=-\zeta\omega_n-i\omega_d}$$

$$c(t) = A\, e^{(-\zeta\omega_n+i\omega_d)t} + B\, e^{(-\zeta\omega_n-i\omega_d)t}$$

$$= e^{-\zeta\omega_n t}\left(A e^{i\omega_d t} + B e^{-i\omega_d t} \right)$$

$$= \alpha\, e^{-\zeta\omega_n t} \cos\omega_d t$$



$$\begin{cases} e^{i\theta} = \cos\theta + i\sin\theta \\ e^{-i\theta} = \cos\theta - i\sin\theta \end{cases} \rightarrow \frac{e^{i\theta}+e^{-i\theta}}{2} = \cos\theta$$

case II) $\zeta = 1$; $s_{1,2} = -\zeta\omega_n$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{(s+\zeta\omega_n)^2} = \frac{\hat{A}}{s+\zeta\omega_n} + \frac{\hat{B}}{(s+\zeta\omega_n)^2}$$

$$\hat{B} = (s+\zeta\omega_n)^2 \left(\frac{\omega_n^2}{(s+\zeta\omega_n)^2} \right)\Big|_{s=-\zeta\omega_n}$$

$$\hat{A} = \frac{d}{ds}\left(\frac{\omega_n^2}{s+\zeta\omega_n} \right)\Big|_{s=-\zeta\omega_n}$$

critical damping

$$c(t) = A e^{-\zeta\omega_n t} + \hat{B} t e^{-\zeta\omega_n t}$$

$$= e^{-\zeta\omega_n t}(\hat{A} + \hat{B}t)$$

III) $(\zeta > 1)$

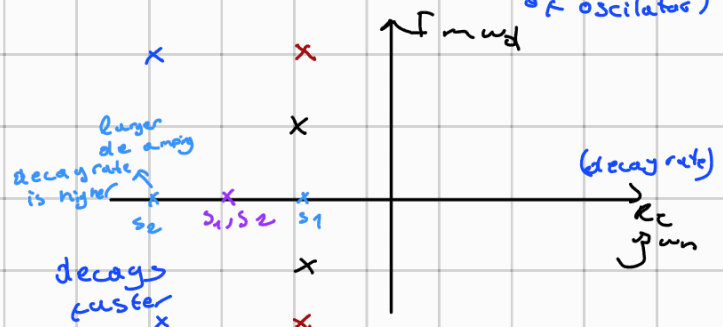
$$c(t) = \hat{A}\, e^{(-\zeta-\sqrt{\zeta^2-1})\omega_n t} + \hat{B}\, e^{(-\zeta+\sqrt{\zeta^2-1})\omega_n t}$$

2nd order systems

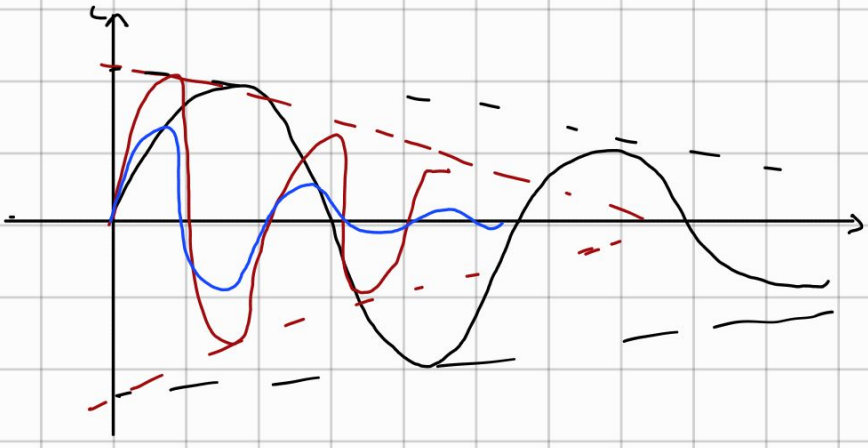
$\zeta < 1$ $\quad s_{1,2} = -\zeta\omega_n \pm i\omega_d$ (Re, Im)

$\zeta = 1$ $\quad s_{1,2} = -\zeta\omega_n$ repeated Re

$\zeta > 1$ $\quad s_{1,2} = -\zeta\omega_n \pm \sqrt{\zeta^2-1}$ Re distinct



If poles are C C → response is oscillatory and decaying



for x $w_d = 0$ no oscilating frequeno

decodc al thesametime

for $\zeta > 1$

$s_1 = -\zeta w_n + \sqrt{\zeta^2 - 1}$

$s_2 = -\zeta w_n - \sqrt{\zeta^2 - 1}$

$C(t) = \underbrace{A e^{s_1 t}}_{\text{this is the most important part}} + B e^{s_2 t}$ → decay fastly

Step response calculation

(just for $\zeta < 1$)

$$\begin{cases} \dfrac{C(s)}{R(s)} = \dfrac{w_n^2}{s^2 + 2\zeta w_n s + w_n^2} \\ R(s) = \dfrac{1}{s} \end{cases}$$

$$C(s) = \frac{w_n^2}{s(s^2 + 2\zeta w_n s + w_n^2)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 2\zeta w_n s + w_n^2}$$

$$= \frac{1}{s} - \frac{s + 2\zeta w_n}{s^2 + 2\zeta w_n s + w_n^2} = \frac{1}{s} - \frac{s + \zeta w_n}{(s + \zeta w_n)^2 + w_d^2} - \frac{\zeta w_n}{(s + \zeta w_n)^2 + w_d^2}$$

$\frac{\zeta}{\sqrt{1-\zeta^2}} \cdot \frac{w_d}{\sqrt{1-\zeta^2}}$ $\quad + \frac{\sqrt{1-\zeta^2}}{\sqrt{1-\zeta^2}}$

$$\mathcal{L}\left\{ e^{-at} \sin \omega t \right] = \frac{\omega}{(s+a)^2 + \omega^2}$$

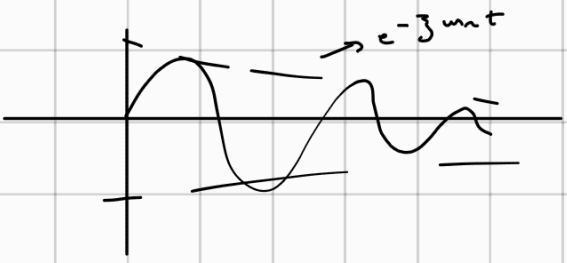
$$\mathcal{L}\left[e^{-a t} \sin \omega t \right] = \frac{s + a}{(s+a)^2 + \omega^2}$$

$w_d = w_n \sqrt{1 - \zeta^2}$

$$c(t) = \mathcal{L}^{-1}[C(s)] = 1 - e^{-\zeta\omega_n t}\cos\omega_d t - \frac{\zeta e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin\omega_d t$$

$$= 1 - e^{-\zeta\omega_n t}\left(\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta}}\sin\omega_d t\right)$$

$$c(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin\left(\omega_d t + \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}\right)$$



$$e(t) = r(t) - c(t) = \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin(\omega_d t + \phi)$$